

GROWRIDER · DESIGN DECISIONS REGISTER

GrowRider design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Amish accepted the recommendations of open items 1 to 4 and 8 to 11 (GRR-DDR-003 accepted); moved to decisions made; open items renumbered 1 to 3
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from REVIEW.md, GRR-DDR-001 to 003 and the build plan work

GrowRider design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md` , GRR-BLD-001) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Helmet: whether one is supplied with each bike (BOM line 20)	Supply with each bike; leave to families or schools; supply through the partner program	None made; a co-design question	Not part of the bike build	GRR-DDR-001, O1
2	First partner (a distribution NGO, a rural school network or a regional assembler) and co-design partner for fit measurements and the regional parts list	To be picked per area, as Amish directed for community designs	None yet	Parts confirmation (items 1 to 12 below) and fit trials	GRR-DDR-001, O2 and O4
3	First region	Zambia, Kenya or Malawi were suggested	None made	Which regional parts are stocked	GRR-DDR-001, O3

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The coaster brake hub is 110 mm over the locknuts with its sprocket on a 42 mm chain line, and its brake ratio (hub torque to sprocket torque)	The dropouts are 110 mm apart and the chainstays are placed for a 42 mm chain line; braking for the smallest rider rests on an assumed ratio of 2.5	GRR-DDR-003, P1 and P16; GRR-DDR-002, D9
2	A long-reach side-pull caliper whose reach covers about 84 mm is stocked. If none is, the fallback is a fork with brake bosses and a short-pull V-brake for children's levers, which is a design decision	The crown height and tire size set the reach	GRR-DDR-003, P14
3	The fork: crown no deeper than 16 mm, 1 in threaded steerer 25.4 x 1.6 mm with 220 mm usable above the crown, slotted fork ends 100 mm apart with eyelets, crown hole for the caliper bolt	The front fender passes 7 mm under the crown; the quill needs the full steerer length	GRR-DDR-003, P2 and P13
4	The solid tire's real width is 50 mm or less on the rim	The rear tire clears the chainstays by 4 mm at a 48 mm width	GRR-DDR-003, P16
5	A 29.8 mm clamp collar closes firmly on the 29.2 mm sleeve, or one is turned to 29.2 mm	Sleeve collars are not a stock size	GRR-DDR-003, P8
6	The 22.0 x 2.0 mm chromoly tube slides in the steerer bore with about 0.2 mm clearance	Steerer bores vary by maker	GRR-DDR-003, P5
7	The 29.2 mm sleeve slides in the seat tube after reaming to 29.4 mm, and the 25.4 mm post in the sleeve	Drawn tube bores vary by about 0.1 mm	GRR-CAL-001, A
8	A 22.2 mm roadster-style bar of about 205 mm sweep and 15 mm rise is stocked	The grips sit where the fit calculation puts them only with this sweep and rise	GRR-DDR-003, P6
9	1 in JIS press-in headset (30.0 mm cups) and a 68 mm BSA bottom bracket are the regional standard	The head tube bore and BB shell are made to them	GRR-DDR-003, P4
10	A builder who can TIG weld 6061 aluminium for the rack, or a bought rack of the same size and rating	Welded aluminium needs skill and has low fatigue strength	GRR-DDR-002, D7
11	Masses of the tires, rims, hub and bought fittings	The mass estimate uses catalog values	GRR-CAL-001, D
12	Which parts are actually stocked in the first region, including coaster hubs and 20 in solid tires	Parts commonality (R10)	GRR-REQ-001, R10

■ Value engineering

Value-engineering target: USD 300 (a hypothetical control target, not a limit; Amish, 2026-10-01: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"). Estimated cost of the constructable design: USD 299 for the bike, BOM lines 1 to 19 (USD 1 under the target), and USD 311 with the helmet (USD 11 over the target). The concept was USD 281 and 293; the parts added for construction account for the USD 18 difference. Frame builder labor is not included.

Main cost drivers:

- Frame tube, plate and consumables, USD 64 (line 1).
- Solid tires, USD 32 for the pair (line 9).
- Rear wheel with coaster hub, USD 30 (line 8).
- Quill stem with its head plate and clamp block, USD 20 (line 5).
- Fork with a long steerer, USD 18 (line 2).

Savings worth trying:

- Thorn-resistant tubes with liners instead of solid tires: about USD 10 and 0.5 kg less, but it reverses D4 (GRR-DDR-001), so it needs Amish's decision.
- A bought steel rack of the right size and rating instead of the welded aluminium one: no aluminium welding skill needed; check mass and price.
- One collar size: a 30.0 mm sleeve would take a stock collar, but the sleeve size was decided (GRR-DDR-002, D10), so only worth revisiting with the strength screen.
- A bought long quill stem cut and fitted with the head plate, if a 22.0 mm chromoly quill of 240 mm is stocked.
- Price the frame kit (tubes, dropouts, BB shell) as one order from a frame-building supplier.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	D1 to D6: 20 in (ISO 406) wheels; coaster brake plus front rim brake; step-through frame with twin down tubes; solid or airless tires for the prototype; one 140 mm crank length; USD 120 production cost target at volume	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	GRR-DDR-001
2026-09-25	D7 to D13: chromoly main tubes, aluminium rack, alloy rims and bar (R5 kept as the production goal); alloy front rim; obtain the coaster hub brake ratio; chromoly steerer and quill, 1.8 mm sleeve; R2 widened to 400 to 670 mm; R9 split into parent and mechanic tool lists; low bar as a co-design question	Amish: "i accept all your recommendations, go with them across all repos."	GRR-DDR-002
2026-09-26	N1: prototype budget raised from USD 250 to USD 300	Amish: "I am ok with the budget top ups"	GRR-DDR-002
2026-09-30	Make the design physically buildable while drawing the build plan; keep open decisions out of the build plan and in this register	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." The changes themselves were accepted on 2026-10-01 (below)	GRR-DDR-003
2026-10-01	budget_usd is a value-engineering target, not a limit; cost is reported over or under it	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register, Value engineering
2026-10-01	Design for construction accepted: the changes P1 to P17 and their knock-on changes, as made	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GRR-DDR-003, Tables 1 and 2
2026-10-01	Quill stem: no positive stop for the prototype; the minimum insertion mark, checked at every fit change, with a bright painted band above it. Follow-up: decide on the stop collar of option (b) after the fit trials	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GRR-DDR-003, A1
2026-10-01	Mass of 15.1 kg accepted for the prototype (R5 production goal of 13 kg still not met); the prototype is weighed at TRL 4	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GRR-DDR-003, A2
2026-10-01	Steerer: keep the 25.4 x 1.6 mm steerer and let the ISO 4210-2 fork test decide	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GRR-DDR-002, N2; GRR-DDR-003, A3
2026-10-01	Renders: 1.30 m rider setting in the product renders, 1.38 m in the concept sheet	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	docs/ REVIEW.md , 2026-09-26, item 2

Date	Decision	Decided by	Record
2026-10-01	Frame color: kit teal for the portfolio renders; the production color is settled in co-design, with a high-visibility option	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	docs/ REVIEW.md , 2026-09-26, item 3
2026-10-01	Painted height scales on the exposed sleeve and quill: a band every 20 mm, a longer band every 60 mm	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	docs/ REVIEW.md , 2026-09-26, item 4
2026-10-01	Markings: the name "GrowRider" screen printed on the chainguard; no head badge	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	docs/ REVIEW.md , 2026-09-26, item 6